

2021

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section-A

(Compulsory)

1. Pick up the correct alternative for each of the following questions : 10×2=20

(a) Which of the following connecting devices typically work at the network layer of the OSI model ?

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(1)

(Turn over)

- (i) Routers
- (ii) Bridges
- (iii) Repeaters
- (iv) Gateway
- (b) Which layer is immediately below the transport layer ?
- (i) Physical
- (ii) Network
- (iii) Application
- (iv) Session
- (c) As data packets moves from the upper to the lower layers, headers are :
- (i) Added
- (ii) Subtracted
- (iii) Rearranged
- (iv) Modified

- (d) The packet of information at the application layer is called :

- (i) Packet
- (ii) Message
- (iii) Segment
- (iv) Frame

- (e) The physical layer concerns with :

- (i) Bits
- (ii) Frames
- (iii) Packets
- (iv) None of these

- (f) A set of rules that governs data communication :

- (i) Protocols
- (ii) Standards
- (iii) Simplex
- (iv) None of the above

(g) The receiver of the data controls the amount of data that are to be sent by the sender is referred to as :

- ☒ (i) Flow control
- (ii) Error control
- (iii) Congestion control
- (iv) Error detection

(h) The pattern of interconnection of nodes in a network is called :

- ☒ (i) Topology
- (ii) Arrangements
- (iii) Connection
- (iv) None of these

(i) Both stations can transmit and receive simultaneously :

- (i) Simplex
- (ii) Half duplex

☒ (iii) Full duplex

(iv) None of these

(j) The connection oriented transport layer protocol is :

☒ (i) UDP

☒ (ii) TCP

(iii) FTP

(iv) NVT

Section-B

Answer any four questions : 4×5=20

- ① • 2. Differentiate between analog and digital signal.
- ③ • 3. What is the purpose of Bridge ?
4. Explain packet switching technique.
- ③ • 5. Explain frequency division multiplexing.
6. The message 11001001 is to be transmitted using CRC polynomial $x^3 + 1$ to protect it from

errors. Calculate the message that should be transmitted.

- Q.7. Define the Repeater. Why are Repeaters needed ?

Section-C

Answer any two questions of the following :

2×15=30

- Q.8. Explain the types of network topology with suitable diagram.
- Q.9. Explain OSI reference model with diagram.
10. Explain stop and wait protocol.
11. Explain the shortest path routing algorithm.

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